📘 Chapter 36-QoS (Quality of Service) 🚀

🌐 What is QoS (Quality of Service)?

QoS is a **network technique** used to:

- Control traffic 🚦
- Prioritize important data 📊
- Manage congestion 🧠

👉 **Simple Meaning:**
"QoS makes sure important traffic (like voice) gets fast delivery."

❓ Why QoS is Needed?

Because not all traffic is equal:

- 📞 Voice needs **fast delivery**
- 🎥 Video needs **smooth streaming**
- 🌍 Web/downloads can tolerate delay

👉 Without QoS:

- Voice becomes **choppy** 😵
- Calls break 📉
- Delay increases ⏱️

✅ QoS Ensures

- 🎯 Voice gets priority
- 📉 Less important traffic gets lower priority
- 🚫 Congestion doesn't break calls

⚙️ QoS Handles These 4 Things

1. **Bandwidth Allocation** 📦
2. **Prioritization** 🥇
3. **Traffic Shaping** 🌊
4. **Queuing** 🧾

🚗 Real-Life Example

Imagine a highway:

- 🚑 Ambulance = Voice (High priority)
- 🚗 Cars = Normal data
- 🚚 Trucks = Background traffic

👉 QoS = Traffic police controlling lanes 🚦

🔷 QoS Types – CoS & DSCP

🟢 1. CoS (Class of Service) – Layer 2

📍 Where Used:

- Inside **Ethernet Frame**
- Works on **Switches**

⚙️ How It Works:

- Uses **3-bit value (0–7)** → called **PCP (Priority Code Point)**

📊 CoS Values Table

CoS	Traffic Type
0	Best Effort
1	Background
2	Excellent Effort
3	Critical Apps
4	Video
5	Voice
6	Internetwork Control
7	Network Control

🧠 Key Points:

- CoS works only with **802.1Q VLAN tagging**
- Found in **Ethernet Header**

📦 802.1Q Tag Format

- TPID → 16 bits
- PCP → 3 bits
- DEI → 1 bit
- VLAN ID → 12 bits

💡 Simple Explanation:

👉 "CoS is a priority label for Ethernet frames."

🔵 2. DSCP (Differentiated Services Code Point) – Layer 3

📍 **Where Used:**

- IP Packets 📡
- Routers 🌍
- WAN / Internet

⚙️ **How It Works:**

- Uses **6-bit value (0–63)**

📊 **DSCP Values**

Traffic	DSCP	Meaning
Voice	EF (46)	Highest Priority
Video	AF41 (34)	High Priority
Normal Data	0	Best Effort
Background	CS1 (8)	Low Priority

💡 **Simple Explanation:**

👉 "DSCP is a priority label for IP packets."

🔥 CoS vs DSCP Summary

Feature	CoS	DSCP
Layer	Layer 2	Layer 3
Device	Switch	Router
Use	Ethernet Frames	IP Packets
Example	CoS 5 (Voice)	DSCP 46

🧠 QoS Core Concepts

🟡 1. Classification

✔ **Meaning:**

Identifying traffic types

🔍 **Based On:**

- IP address
- Ports (e.g., SIP 5060)
- Protocol
- VLAN
- MAC

👉 **Simple:** Sorting packets 📦

🟢 2. Marking

✔ **Meaning:**

Assigning priority label

- CoS → Layer 2
- DSCP → Layer 3

👉 **Simple:** Putting a sticker 🏷️

🔴 **3. Policing**

✔️ **Meaning:**

Limits traffic rate

- Extra traffic → ❌ dropped
- Or remark lower priority

👉 **Simple:** Speed limit 🚓

🔵 **4. Shaping**

✔️ **Meaning:**

Smooth traffic

- Extra packets → stored (not dropped)

👉 **Simple:** Traffic control 🚦

🟣 **5. Queuing**

✔️ **Meaning:**

Packets placed in different queues

- High → Voice 📞
- Medium → Video 🎥
- Low → Data 🌍

👉 **Simple:** VIP line 🎟️

⚙️ 6. Scheduling

Decides:

- Which queue goes first 🥇
- How much bandwidth each gets

📊 Queuing Methods

1️⃣ FIFO (First In First Out)

- No priority ❌
  👉 "First come, first served"

2️⃣ Priority Queuing (PQ)

- High priority first 🥇
- Low may starve 😢

👉 "VIP goes first"

3️⃣ WFQ (Weighted Fair Queuing)

- Fair sharing ⚖️
- Important gets more

👉 "Everyone gets share"

4️⃣ CBWFQ

- Custom classes
- Bandwidth allocation

👉 "Admin decides traffic rules"

5️⃣ LLQ (Low Latency Queuing)

- Strict priority for voice 📞
- Reduces delay & jitter

⚠️ Problem:

- Can starve others

👉 **Solution:** Use policing

📉 Congestion & Packet Handling

🚨 **Problem:**

When traffic > bandwidth
👉 Queue fills → packets drop

❌ **Tail Drop:**

- Drops packets when full

🧠 **RED / WRED:**

- Drop packets early to avoid congestion

🧠 Advanced QoS Concepts

🏷️ **IP ToS Byte**

- 6 bits → DSCP
- 2 bits → ECN

🧓 IP Precedence (Old)

- Similar to CoS
- Example:
  - 5 → Voice
  - 4 → Video
  - 0 → Best effort

🚀 Modern DSCP Standards

✔ **DF (Default Forwarding)**

- Value = 0
- Normal traffic

✔ **EF (Expedited Forwarding)**

- Value = 46
- Voice traffic 📞

✔ **AF (Assured Forwarding)**

- 4 classes
- Each has 3 drop levels

Example:

- AF41 → Best
- AF13 → Worst

📜 RFC 4954 Recommendations

- Voice → EF
- Video → AF4x
- Streaming → AF3x
- High data → AF2x

- Best effort → DF

🔐 Trust Boundary

✔️ **Meaning:**

Where network decides:

- Trust markings ✅
- Or overwrite ❌

📌 **Example:**

- IP Phone → Trusted ✅
- PC → Not trusted ❌

👉 Prevents users from cheating priority 🎭

📦 **Traffic Flow (End-to-End)**

1. Classification 🔍
2. Marking 🏷️
3. Queuing 🧾
4. Scheduling ⚙️
5. Policing/Shaping 🚦

⚙️ QoS Commands (Important) 💻

🎧 1. Voice VLAN

🔧 Command Set

```
interface fa0/1
switchport mode access
switchport access vlan 10
switchport voice vlan 20
mls qos trust cos
```

📖 Explanation

🛠️ Command	💡 Meaning
switchport mode access	🔹 Port ko access port banata hai (trunk nahi hota).
switchport access vlan 10	💻 PC se aane wala **data traffic VLAN 10** me jata hai.
switchport voice vlan 20	📞 IP Phone ka **voice traffic VLAN 20** me jata hai (better security & QoS).
mls qos trust cos	⭐ Switch CoS value ko trust karta hai (IP phone priority mark karta hai aur switch usko follow karta hai).

🎯 2. CoS (Layer 2 Priority)

🔧 Commands

```
mls qos trust cos
mls qos cos 5
```

📖 **Explanation**

🔹 **mls qos trust cos**

- Switch Ethernet frame ke **CoS value ko trust karta hai** ✅
- IP Phones usually **CoS = 5** send karte hain 📞
- Isse voice traffic ko **high priority milti hai** 🚀

🔹 **mls qos cos 5**

- Agar device khud marking nahi karta ❌
- To switch forcefully **CoS = 5 assign karta hai** ✔️

🌐 3. DSCP (Layer 3 Priority)

🔧 **Commands**

```
mls qos trust dscp

class-map match-all VOICE
match ip dscp ef
```

📖 **Explanation**

🔹 **mls qos trust dscp**

- Switch device se aane wali **DSCP values ko accept karta hai** ✔️

🔹 **class-map match-all VOICE**

- 📦 "VOICE" naam ka **traffic group create karta hai**

🔹 **match ip dscp ef**

- 🎙️ DSCP EF (46) packets ko match karta hai
- Ye standard hai **VoIP RTP traffic ke liye**
- Network devices easily **voice traffic identify kar paate hain**

⚙️ 4. QoS Methods Explained with Commands

🅰️ A) Classification Commands – Detailed

class-map match-any VOICE
match ip dscp ef

📖 Explanation

- 🧠 Class-map ka use **traffic identify karne ke liye hota hai**
- Ye **DSCP EF (46)** wale packets select karta hai
- 🎧 Ye packets mostly **VoIP audio streams** hote hain
- 🚦 **Classification QoS ka first step hota hai**

🅱️ B) Marking Commands – Detailed

policy-map MARKING
class VOICE
set dscp ef

📖 Explanation

- 🛠️ Policy-map class par **actions apply karta hai**
- set dscp ef → packets ko **high priority mark karta hai**
- 🎯 Goal: Voice ko **poore network me priority mile**

🅲 C) Policing Commands – Detailed

policy-map POLICE
class VOICE
police 8000
conform-action transmit
exceed-action drop

📖 **Explanation**

- ⏱️ police 8000 → allowed rate = **8000 bits/sec**
- ✔️ Limit ke andar → **transmit**
- ❌ Limit exceed → **drop**
- 🛡️ Policing greedy applications ko **network flood karne se rokta hai**

🅳 D) Shaping Commands – Detailed

policy-map SHAPE
class class-default
shape average 2000000

📖 **Explanation**

- 📉 shape average 2000000 → traffic ko **2 Mbps tak control karta hai**
- 🚦 Traffic drop nahi hota, **smooth/delay hota hai**
- 🌍 WAN links par useful hai jahan burst se packet loss hota hai

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🔍 6. Verification Commands – Detailed

✅ Check if policy is working

show policy-map interface

📊 **Shows:**

- Kaunsa class match ho raha hai
- Kitna traffic aa raha hai
- Drops, bandwidth, queuing details

✅ **Check interface QoS trust**

show mls qos interface

📊 **Shows:**

- Interface **CoS ya DSCP trust karta hai ya nahi**

✅ **Check DSCP and CoS values**

show mls qos maps dscp-out
show mls qos maps cos-out

📊 **Shows:**

- 🔄 CoS ↔ DSCP mapping

🧠 **Final Easy Summary** 🧾

Feature	Meaning
Classification	Identify traffic
Marking	Add priority
Policing	Drop extra traffic
Shaping	Smooth traffic
Queuing	Put in lines
Scheduling	Decide order

🏁 **Super Simple Final Line**

👉 **QoS = Smart traffic control system that ensures voice and important data always get priority.** 🚀 📞

🧠 ✅ 1️⃣ SUMMARY

🌐 What is QoS?

👉 QoS = **Traffic Control System**

✔ Controls traffic 🚦
✔ Prioritizes important packets 🥇
✔ Manages congestion 📉

💡 Simple:
👉 "Important traffic goes first"

❓ Why QoS Needed?

Without QoS ❌:

- Voice breaks 📞 💥
- Video lags 🎥
- Delay increases ⏱️

With QoS ✅:

- Voice = smooth
- Video = stable
- Data = controlled

⚙️ QoS Handles

1️⃣ Bandwidth 📦
2️⃣ Prioritization 🥇
3️⃣ Shaping 🌊
4️⃣ Queuing 🧾

🔹 CoS vs DSCP

Feature	CoS	DSCP
Layer	L2	L3
Bits	3 bits	6 bits
Device	Switch	Router
Example	CoS 5	DSCP 46

✓ CoS → VLAN tag
✓ DSCP → IP packet

📊 Important Values

- Voice → **EF (46)** 📞
- Video → **AF41** 🎥
- Data → **0 (Best Effort)**

🧠 QoS Core Steps

1️⃣ Classification 🔍
2️⃣ Marking 🏷️
3️⃣ Policing 🚓
4️⃣ Shaping 🚦
5️⃣ Queuing 🧾
6️⃣ Scheduling 🥇

📊 Queuing Types

Type	Meaning
FIFO	No priority
PQ	Voice first

Type	Meaning
WFQ	Fair
CBWFQ	Custom
LLQ	Voice priority

📉 **Congestion Handling**

- Tail Drop → Drop when full ❌
- RED/WRED → Smart early drop 🧠

🔐 **Trust Boundary**

✔ Trust IP Phone
❌ Don't trust PC

👉 Prevents fake priority marking

🎯 ✅ 2️⃣ CONCLUSION

👉 QoS is **mandatory in modern networks** because all traffic (voice, video, data) shares same infrastructure.

✔ Ensures:

- Low delay
- Low jitter
- Low packet loss

✔ Key idea:
👉 **Voice always gets highest priority**

💡 Final Line:
👉 "QoS is a smart traffic control system that ensures business-critical applications run smoothly."

💥 🎯 3️⃣ Q & A

❓ Q1. What is QoS?

✅ **Answer:**

QoS (Quality of Service) is a networking technique used to prioritize important traffic and manage congestion.

👉 It ensures:

- Voice/video gets priority
- Network performance remains stable

❓ Q2. Why QoS is required?

✅ **Answer:**

Because all traffic shares same network.

Without QoS:

- Voice breaks
- Delay increases

With QoS:

- Critical traffic gets priority

❓ Q3. What are QoS goals?

✅ **Answer:**

- Reduce delay
- Reduce jitter
- Reduce packet loss
- Improve bandwidth usage

❓ **Q4. What is CoS?**

✅ **Answer:**

CoS (Class of Service) is a Layer 2 priority marking inside Ethernet frame.

✓ Uses 3 bits (0–7)

❓ **Q5. What is DSCP?**

✅ **Answer:**

DSCP is a Layer 3 priority marking in IP header.

✓ Uses 6 bits (0–63)

❓ **Q6. Difference between CoS and DSCP?**

👉 Important answer:

- CoS → Switch (Layer 2)
- DSCP → Router (Layer 3)
- DSCP is more scalable

❓ **Q7. What is EF?**

✅ **Answer:**

EF (Expedited Forwarding) = DSCP 46

✓ Used for voice traffic
✓ Highest priority

❓ **Q8. What is AF?**

✅ **Answer:**

AF (Assured Forwarding) is used for different priority levels.

Example:

- AF41 → high priority

❓ **Q9. What is Classification?**

✅ **Answer:**

Identifying traffic type.

👉 Based on:

- IP
- Port
- Protocol

❓ **Q10. What is Marking?**

✅ **Answer:**

Assigning priority to packets.

👉 Example:

- DSCP EF for voice

❓ **Q11. What is Policing?**

✅ **Answer:**

Limiting traffic rate.

✓ Extra traffic → dropped

❓ **Q12. What is Shaping?**

✅ **Answer:**

Delays extra traffic instead of dropping.

✓ Smooth traffic flow

❓ **Q13. Difference between Policing and Shaping?**

Feature	Policing	Shaping
Action	Drop	Delay
Use	Security	WAN

❓ **Q14. What is Queuing?**

✅ **Answer:**

Placing packets in different queues based on priority.

❓ **Q15. What is Scheduling?**

✅ **Answer:**

Deciding which queue is served first.

❓ **Q17. What is congestion?**

✅ **Answer:**

When traffic exceeds bandwidth.

❓ **Q18. What is Tail Drop?**

✅ **Answer:**

Packets dropped when queue is full.

❓ **Q19. What is WRED?**

✅ **Answer:**

Drops packets early to avoid congestion.

❓ **Q20. What is Trust Boundary?**

✅ **Answer:**

Point where QoS markings are trusted.

✓ IP Phone → trusted
❌ PC → not trusted

❓ **Q21. Why trust boundary important?**

✅ **Answer:**

Prevents users from assigning fake priority.

🔥 🔹 **SCENARIO QUESTIONS**

❓ **Q22. Voice breaking in network?**

✅ **Answer:**

- No QoS
- High delay
- Packet loss

❓ **Q23. Video buffering?**

✅ **Answer:**

- No bandwidth allocation
- Wrong queuing

❓ **Q24. Users abusing bandwidth?**

✅ **Answer:**

Use:
👉 Policing

❓ **Q25. WAN congestion?**

✅ **Answer:**

Use:
👉 Shaping

❓ **Q26. How to prioritize voice?**

✅ **Answer:**

- Mark DSCP EF
- Use LLQ

❓ **Q27. End-to-end QoS flow?**

✅ **Answer:**

1. Classification
2. Marking
3. Queuing
4. Scheduling
5. Policing/Shaping

🧠 🔥 **One Liner**

👉 "Explain QoS completely"

💯 **Perfect Answer:**

QoS is used to prioritize important traffic like voice and video in a network. It works by classifying traffic, marking it using CoS or DSCP, placing it into queues, and scheduling transmission based on priority. It also uses policing and shaping to control bandwidth and prevent congestion. This ensures low delay, low jitter, and minimal packet loss, which is critical for real-time applications like VoIP.

🚀 **FINAL TAKEAWAY**

✔ QoS = Traffic control system 🚦
✔ CoS = Layer 2 priority
✔ DSCP = Layer 3 priority
✔ LLQ = Best for voice 📞
✔ Policing = Drop
✔ Shaping = Delay
